

Sleeping Machines

Computing with races, cancellations and near misses · status report, 27 September 2026

Headline: a learned event network beats a clocked dense model at $\sim 10^5\times$ lower cost

On a timing task (“B within Δ after A, unless C”, 4 patterns + none, 12 channels), with **nothing given**: each detector learns which channel holds, which triggers, how long it holds and which channels veto, by errors-only local updates (E35).

model	test accuracy	cost per episode	learning
event network (E35)	1.000 $\times 4$, 0.9995 (5 seeds)	7.5 synaptic events	443–1,530 updates in 200k episodes
clocked conv net, best (E32)	0.995	3.07M multiply-adds	backprop, 200k episodes
clocked conv net, cheapest ≥ 0.99	0.992	768k multiply-adds	backprop
clocked conv net at ≈ 0.91	0.895	1.9k multiply-adds	backprop

More accurate than the best dense model at about $10^5\times$ fewer operations; the event cost does not grow with silence, the clocked cost does. Priors on both sides: hold/veto nodes (§64) vs a receptive field matched to the pattern length. It worked because of theory: order is an asymmetry of PSP durations and veto needs the held interval (§61, §64); a node computing one interval predicate needs only $O(\text{its few parameters})$ mistakes. **Depth pays too (E34)**: hold/trigger chains reach 0.97 on hierarchical motifs (0.86 with a generic window bank) vs 0.39 at depth 1, 5 seeds. A Transformer on event tokens is the next baseline; grokking remains open.

In one page

Sleeping Machines proposes that computation can happen **in time rather than memory**: candidate events race, the first to fire cancels the rest, and the cancelled ones keep a trace of how close they came, which a later teaching signal can use. Every experiment below had its predictions written down before evaluation, and the report includes what did not work.

What holds up

- **A cancelled node can be taught (E4).** Keeping a loser's distance to threshold lets a delayed teacher promote the right answer. At $K = 128$ classes: 0.79 accuracy where the reward-modulated rule is at chance, with 6% of the weight updates of uniform credit.
- **Races decide as fast as the evidence allows (E2).** An accumulator race beats a fixed-time decoder at every decision time and tracks the optimal MSPRT, using only additions and a threshold.
- **Learning work tracks activity, not capacity (E5):** about $300\times$ fewer weight updates than a sparse softmax on the same connectivity.
- **Local, event-driven learning reaches about 96% on MNIST (E6, E14).** One hidden layer 95.9% (test, round 3); with credit conservation 96.0% (validation, 2 seeds).
- **Counterfactual credit pays with depth (E14, full length).** Over fired-only credit: +0.2, +1.2, +1.5 points at depths 1–3 (2 seeds), +1.9 and +2.5 at depths 4–5 (1 seed). A frozen hidden stack collapses.
- **Credit conservation, predicted by the theory, holds at full length:** +1.0 to +1.7 points at every depth, both seeds.

What does not (yet)

- **Depth still costs accuracy** (0.960, 0.952, 0.941 at depths 1–3 with conservation). The theory now names three reasons (credit contraction, activity drift, pattern chaos), each with a predicted remedy, all untested.
- **Energy.** Only the single racing layer beats an equally accurate dense model at inference (about $2.4\times$). Dense hidden layers cost more than a small MLP; sparse fan-in ($14\text{--}22\times$ fewer events) is the candidate fix.
- **Leads that reversed:** the shadow neuron (debug +5 points, full length -0.8); the counterfactual routing gradient in MoE (a tie with load balancing at 10 seeds).
- **Spiking Heidelberg Digits (E22), the first event-native benchmark: poor.** Race 0.35–0.36 vs dense MLP 0.56–0.59; a frozen random hidden layer beats trained ones. Later decisions help only slightly; the local learning rule is the main problem there.
- **Residual streams did not rescue depth** under local learning (depth 3: 0.915 delay-matched, 0.898 plain, 0.937 without), at $2\text{--}4\times$ the synaptic events.
- **Market stream (E17): no edge.** The race matches simple baselines while deciding a third earlier, but continual learning did not help, learned trade selection had no skill, and every learner loses money after costs.
- **Not yet run:** the E7 stream learner; most theory predictions (M31–M43).

New on 26–27 September: computing with time (details on the E24–E30 pages)

- **In a race, winning is positional (E26, 3 seeds).** Sparse, error-driven delay learning that pulls only the teacher learns $(a + b) \bmod p$ (0.97–0.98 at 50% of pairs, $p = 31$); pushing the wrong winner stays at chance in every run. Displacing false positives is equally destructive (E27, 5 seeds: 0.195 vs 0.914 with veto).
- **The operator basis is Turing-complete (E30).** A Minsky machine wired from delay, or, and, veto and one oscillator runs exactly; restoring phases once per cycle makes reliability independent of program length.
- **Depth needs counterfactual routing credit (E28),** which cancellation supplies; with it depth is learnable, but it does not beat depth 1 yet: hidden nodes become class detectors, not parts.

- **No true grokking yet (E29).** E25's generalization with delays is restriction by the readout, not grokking; a general network with learned loops has not generalized.
- **A measured frontier (E32).** At matched accuracy (≈ 0.91) on a timing task the event learner uses $\approx 190\times$ fewer operations than the cheapest clocked dense model, $\approx 19,000\times$ inside 99% silence, $\approx 6,000\times$ fewer in training; the dense model can reach 0.995 at far higher cost, and a sparse (event-driven) dense model narrows the gap to $\approx 10\times$.
- **Negative:** pull-only, conserved budgets and prices do not transfer to the main weight race (SHD $0.35 \rightarrow 0.04\text{--}0.29$ across variants), and SHD offers only $\sim 6\times$ input-side advantage at the bins dense models use: not a supremacy benchmark.

What is new in the theory (details in the theory pages)

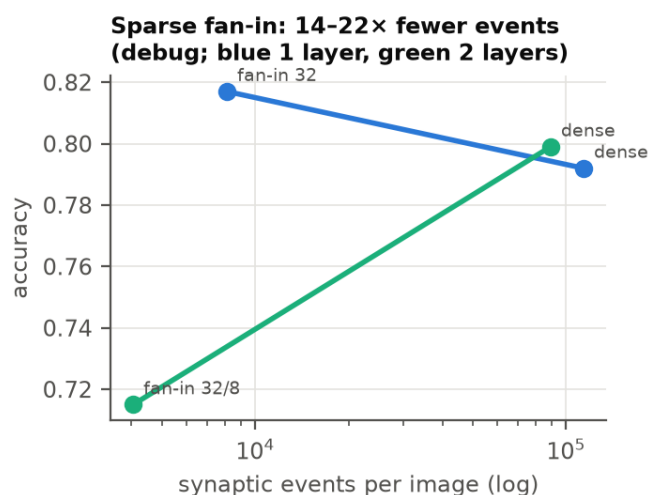
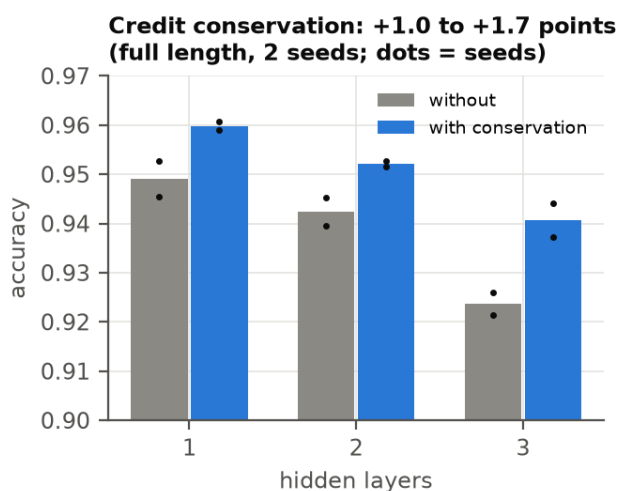
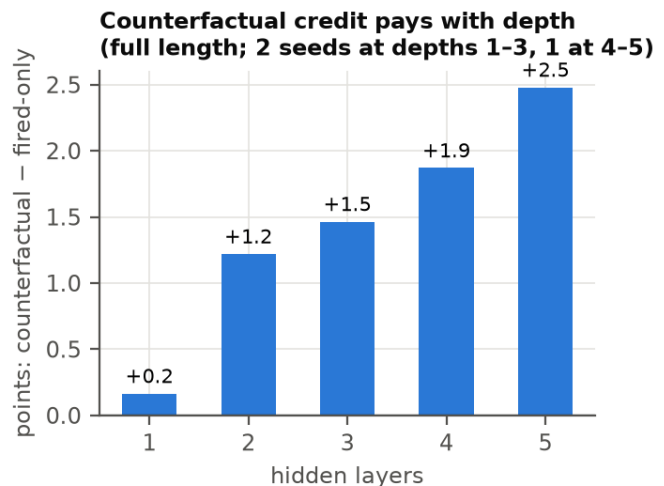
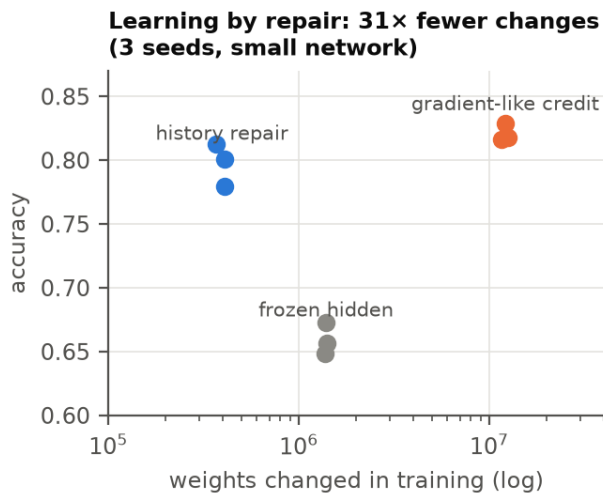
- **Exact results:** timing credit sums to the deadline's credit (a Ward identity); excitatory race networks are topical maps, so timing noise is never amplified and a decision comes with a certified jitter radius; committing to a branch costs temperature $\times$ surprisal, and near-miss credit is the gradient of that cost.
- **Mechanisms, predicted and queued for test:** why deep credit dies (contraction by a Markov kernel; an activity mode that swamps evidence), why prices must be the faster timescale, and why firing *patterns* are chaotic even though firing times are not.
- **Checked against the literature:** several pieces turned out to be prior art and are credited (for example the polyhedral geometry of first-spike networks, the outlier mode in non-negative backprop).

Direction (decided 26 September)

The project's identity is local, sparse, error-gated learning on an asynchronous substrate: learning whose cost follows events and errors, decisions that take as long as the evidence needs, and credit through what did not happen. Matching dense accuracy by training spiking networks with backpropagation is an established field in which this project would only be catching up.

- **Exact-gradient training is a diagnostic ceiling, not the method.** With exact gradients and no cancellation the race architecture reaches 0.9675 vs 0.976 for an MLP (MNIST, depth 2, matched budget): the architecture is close to dense, and the gap is mostly the local learning rule plus ~2 points for cancellation (Fermi-Dirac training recovers ~60% of it).
- **Every result reports its energy side** (synaptic events, spikes, weight updates), judged on the accuracy-vs-energy frontier against dense models.
- **Every mechanism gets a locality audit:** per node is fine; a slow per-layer broadcast is acceptable; a global backward pass is diagnostic only.
- **Benchmarks where asynchrony and continual learning are native:** SHD (E22), class-incremental streams (E23), then NeuroBench's keyword few-shot class-incremental task.
- **Honest assessment of the theory:** mostly known mathematics applied to race networks. It changed results in four places (conservation +1.0–1.7; depth-3 exact training rescued 0.10 → 0.87; delay-matched skips +1.6 over plain; Fermi-Dirac training +1.2). The next genuine step is a result about what local learning can and cannot learn; \$50 (bounded forgetting) is the first aimed at the project's own niche.

Most promising so far

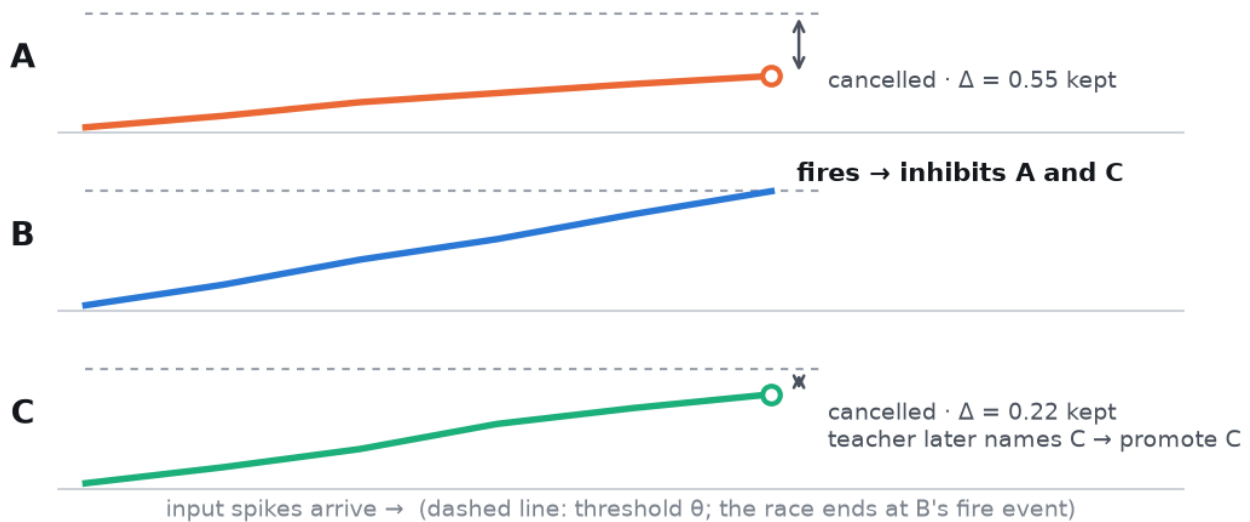


- **Learning by repairing history** (top left, 3 seeds): fix each mistake at its pivotal branch point with the smallest change. Within ~2 points of gradient-like rules while changing **31× fewer weights**, which matters for continual learning (less interference) and for hardware (fewer writes).
- **Counterfactual credit pays with depth** (top right, full length): no gain with one hidden layer, +1.2 points with two and +1.5 with three (2 seeds, both positive), +1.9 with four and +2.5 with five (1 seed): the gap grows monotonically with depth. Near-miss nodes influence the output only through events that did not happen, which fired-only credit cannot see.
- **Credit conservation, confirmed at full length** (bottom left; 2 seeds, width 400, 3 epochs): normalising competitor credit at each race, as the theory's soft-minimum derivative requires, gives 0.960 / 0.952 / 0.941 at depths 1 / 2 / 3 against 0.949 / 0.942 / 0.924 without: +1.0 to +1.7 points at every depth, both seeds. (The debug runs had suggested +8 to +10; full training shrinks it.)
Correction: the shadow neuron's debug lead did *not* hold at full length.
- **Sparse connectivity** (bottom right, debug): 14-22× fewer synaptic events per image. In one layer it even raised accuracy; with two layers it cost accuracy in a very short run. Full runs are queued. For scale, the MLP that matched round-3 accuracy (0.96) needs ~25k multiply-accumulates; sparse race networks at 4-8k events would win at inference if they hold accuracy.
- **Theory that predicts:** reading the network's unrealised futures as a dequantized tropical computation predicted credit conservation (confirmed in debug) and exact time-shift invariance (confirmed); its asynchronous form (shadow events) reproduces batch computation exactly (M20).

Evidence levels: “3 seeds” and “full length” results are confirmatory-grade within their setting; “debug” results are single short runs, reported as leads.

The idea

One race: B fires, A and C are cancelled but keep their distance to threshold

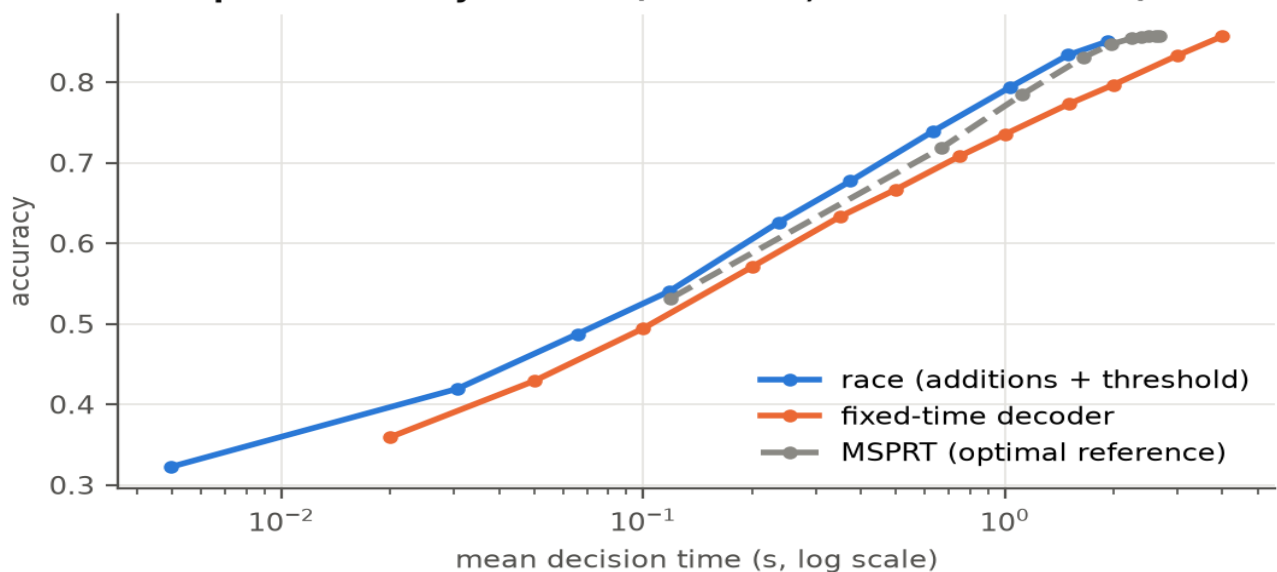


Nodes are non-leaky integrate-to-threshold units. The first to reach threshold fires and cancels the others; each cancelled node freezes Δ , its normalised distance to threshold. A teaching event arriving later can then credit near misses, not only the node that fired. The simulator has no global clock: it jumps from event to event and counts every operation, and all work and energy figures come from those counts.

How a decision unfolds (weaving). Each group of neurons holds an open set of possible futures: every member's projected crossing time, which only moves earlier as input arrives. A crossing is *woven*, fixed history. When k members have crossed, the group closes, and each loser's distance to threshold is frozen as a near miss. Spikes from closed groups drive the next layer, so the settled region spreads through the network as a diagonal front in layers and time, until the first output crossing decides. Learning rereads the woven record, including the near misses. (experiments/WEAVING.md)

E2 · decisions that take as long as they need

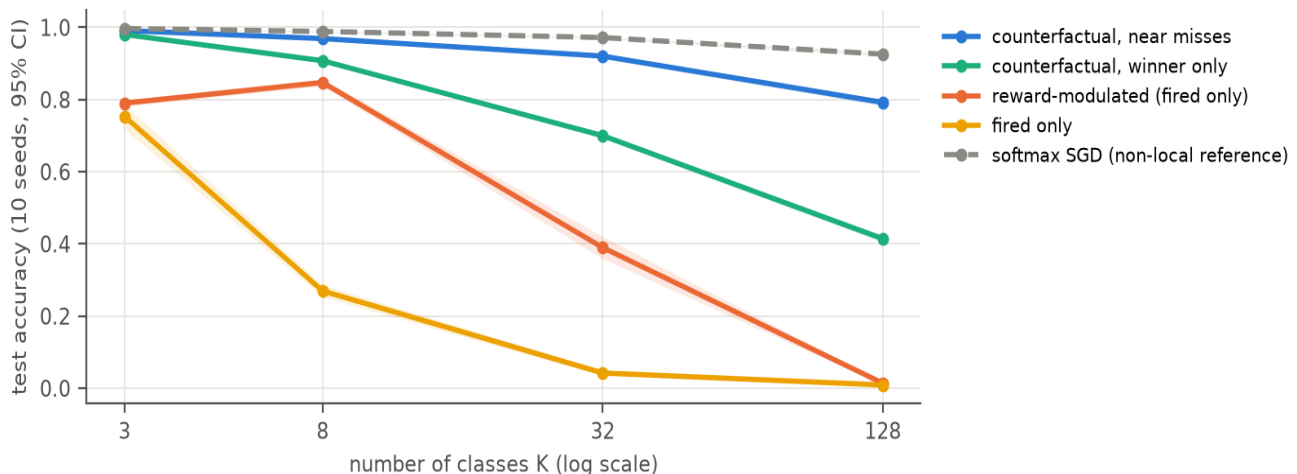
E2 · speed-accuracy frontier (4 classes, Poisson evidence)



The race dominates a fixed-time decoder at every matched decision time and follows the MSPRT, which knows the exact likelihoods. Easy trials end in 0.29 s, hard ones in 1.69 s, with no controller deciding when to stop.

E4 · counterfactual credit

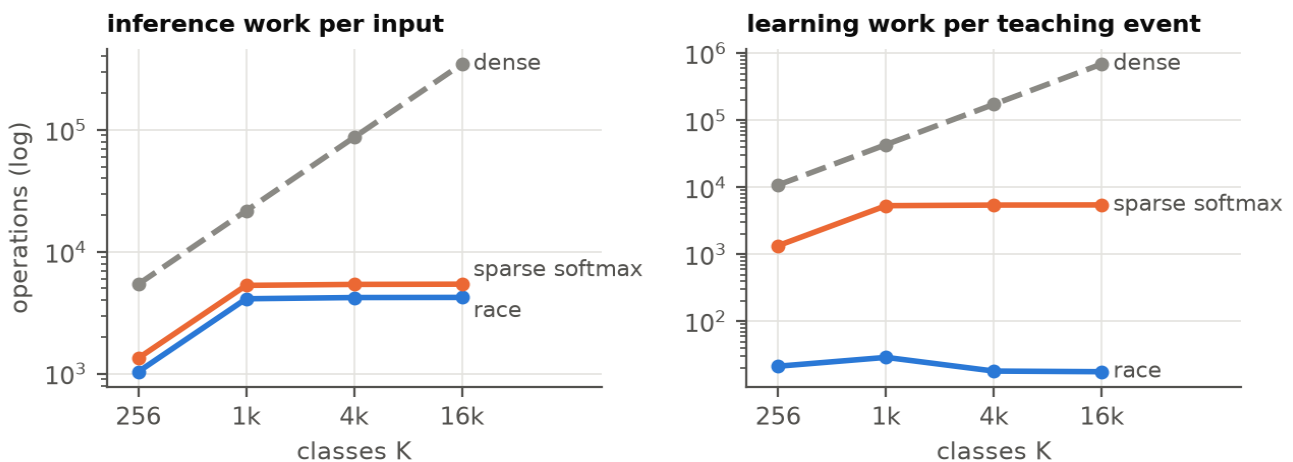
E4 · a cancelled node can be taught



Rules that credit only nodes that fired can only punish a wrong winner; as K grows the right answer almost never fires, and they collapse to chance. Counterfactual rules can promote the target from its frozen record. Crediting only competitors that came close is both the most accurate and the cheapest local rule. It still trails the global-gradient reference by 13 points at K = 128, and credit survives a delayed teacher only until another race overwrites the record.

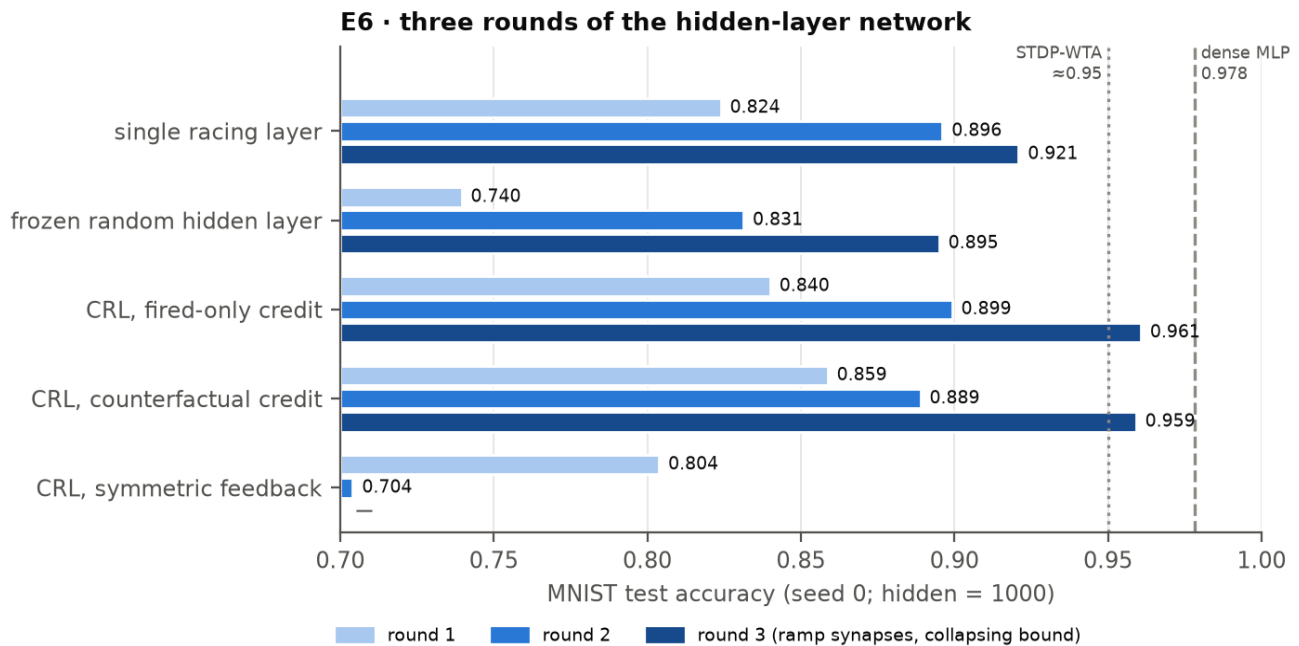
E5 · capacity that costs nothing while it sleeps

E5 round 2 · work as capacity grows 64× (3 seeds)



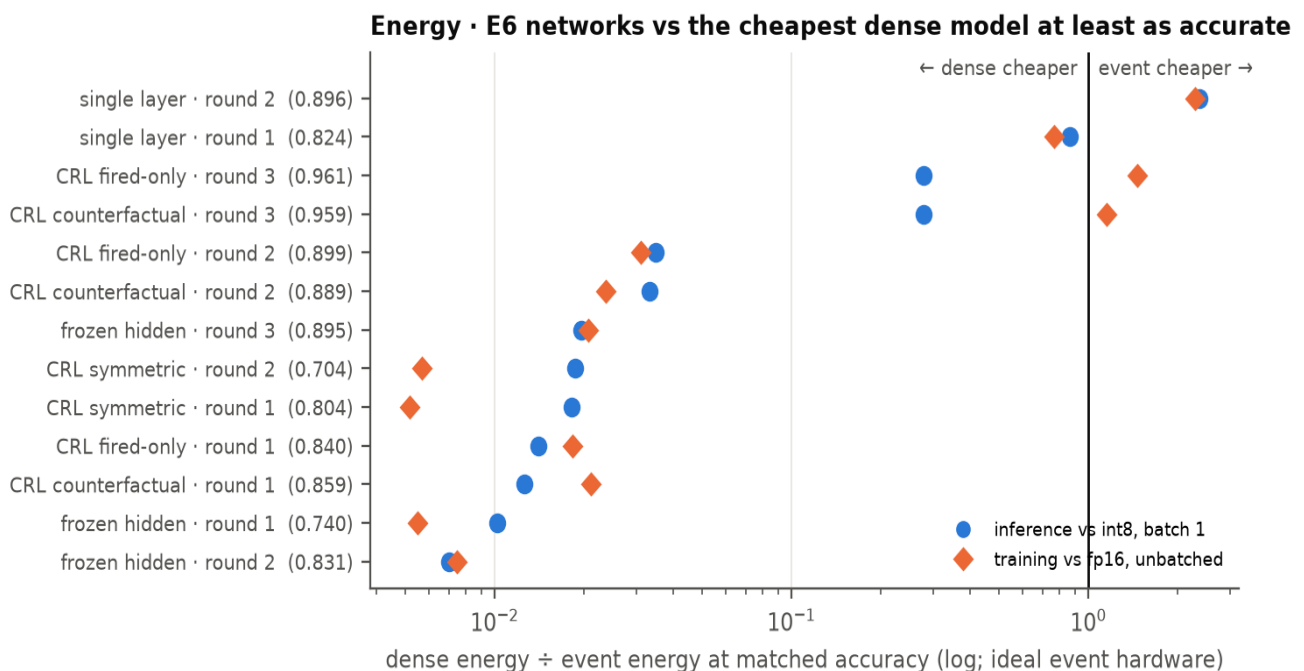
The fair baseline is a sparse softmax on the same connectivity, because most of what event computing saves comes from sparsity itself. Against it, inference work is similar (the race saves about 20% by stopping early), while learning work is about 300× lower: only near misses are updated. Dense models grow linearly with K.

E6 · hidden layers on MNIST



- Round 3 (current-based ramp synapses, a collapsing decision bound, 3 winners per group) lifted the hidden networks from about 0.89–0.90 to about 0.96.
- Hidden learning matters: the frozen random hidden layer stays at 0.895.
- With one hidden layer, fired-only $\approx$ counterfactual credit. From two layers on, counterfactual credit leads, and the gap grows with depth (E14, on the depth page). Symmetric feedback was worst in rounds 1–2 and was not carried forward.
- A wider hidden layer (2000 nodes): 96.0%, no gain. The single-layer control reaches 92.1%: the new synapse model gave ~ 2.5 points, depth ~ 4 more.

Energy, measured by counting

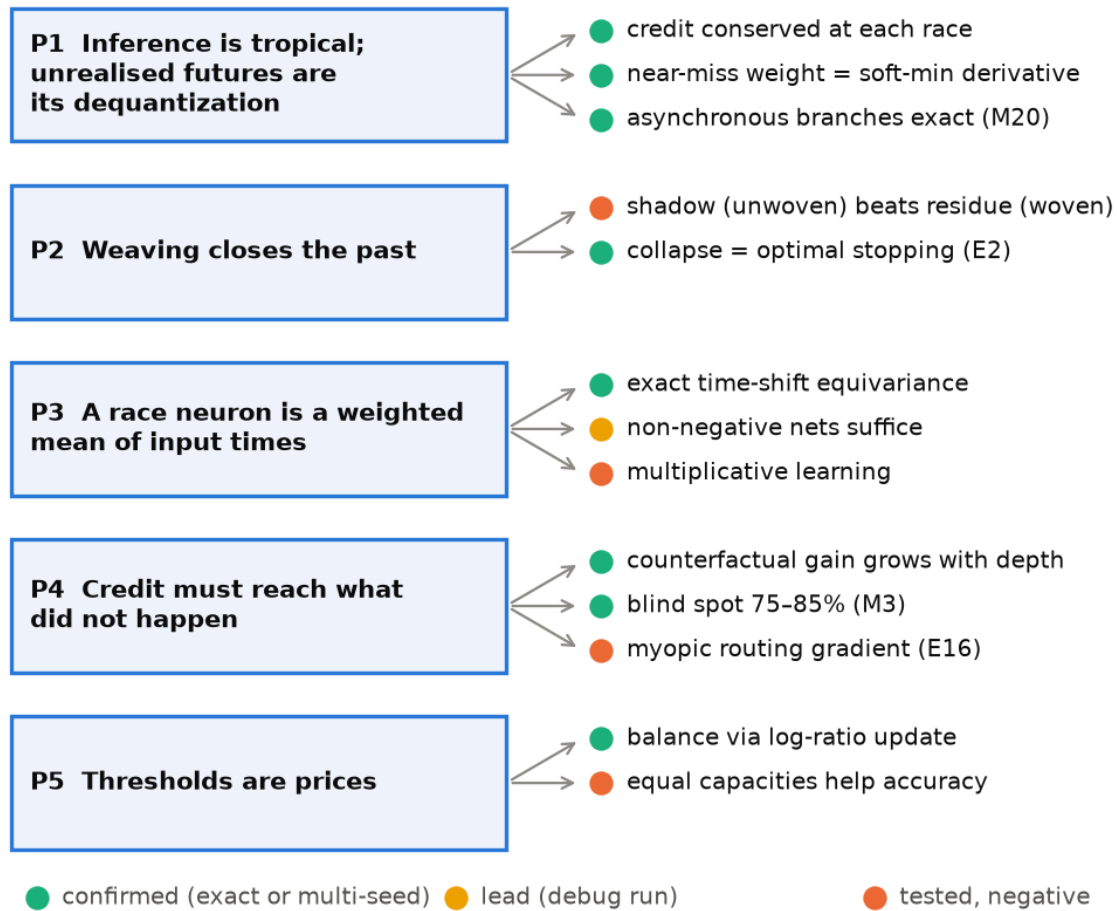


Operation counts priced with published per-operation energies (45 nm logic and SRAM; measured Loihi). These are order-of-magnitude estimates, not chip measurements. At inference, the single racing layer is the only network that beats an equally accurate dense model; the round-3 hidden networks are 1.2–1.5 $\times$ cheaper to *train* than unbatched dense training. Hidden layers integrate about 100 input

events in each of 1000 nodes before inhibition stops them, and that swamps the savings. Against batched dense hardware every advantage disappears.

The theory in five principles

The theory note (experiments/THEORY.md) has grown to some twenty-five sections. They reduce to five principles; each result follows from one of them. Colours give the evidence status.



What the theory added since (THEORY §27-41)

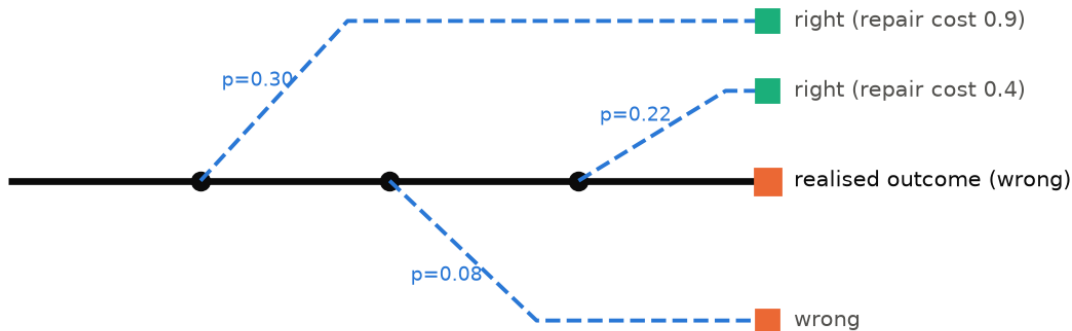
Each result is labelled by what kind of claim it is. **Exact**: a derivation that holds without approximation. **Scaling**: an order-of-magnitude argument whose exponent or sign is the prediction. **Prior art**: found in the literature, credited. None of the predictions below has been confirmed yet; the tests are queued.

result	kind	prediction / status
Timing credit sums to the deadline's credit (Ward identity, §30.1)	exact	centring timing credit is the symmetry, not a heuristic
Excitatory race nets are topical maps: timing noise never amplified; certified jitter radius (§34)	exact	zero flips among certified samples (M36)
Committing to a branch costs $\sigma \times$ surprisal; near-miss credit is its gradient (§35)	exact	supervised loss = cost of weaving the teacher's branch
Deep credit contracts like a Markov chain; exact kernels conserve errors in the sum (§30-31)	scaling	share Jacobian + centring trains depth 3 (M34)
Credit through positive weights collapses to an activity (Perron) mode (§27, §29)	scaling; outlier mode is prior art	Perron centring $\geq$ mean centring (M33)
Prices must be the faster timescale; our default is 10 $\times$ too slow (§36)	scaling	threshold near $\kappa \approx \eta$ for pivotal credit (M38)
Firing patterns are chaotic: $\rho' \approx A\sqrt{\rho}$, no ordered phase; $A \propto 1/\sqrt{k}$ (§41)	scaling	slope $\frac{1}{2}$ in $\log \rho$ across layers (M43)
Winner-fan-in coupling $k \cdot F \geq G$; widths from the data's entropy exponent (§37)	scaling	entropy exponent measured: $\alpha \approx 0.95$, so no pyramid from input redundancy (refuted a guess)
Optimal weaving prices commitment cost (MSPRT); race neurons are blind to absence (§38)	scaling; MSPRT is prior art	relative stopping beats the absolute race (M40)
Concave piecewise-linear firing time, one piece per causal set (§34.1)	prior art	polyhedral geometry of TTFS networks (2026)
Deep exact training collapses; fast homeostasis + Ward centering rescue it (§27, §30, §36)	scaling, confirmed	depth-3 exact training 0.10 $\rightarrow$ 0.87 (debug)
Skips must be delay-matched or the shallow path wins the race (§47)	scaling, partly refuted	0.915 vs 0.898 plain, but no skips best (0.937) under local learning
The entropic k-winner race is Fermi-Dirac; chemical potential = price (§48)	exact; soft top-k prior art	recovers $\sim 60\%$ of the cancellation cost
Dilation equivariance; temporal collapse; temporal normalisation as a gauge (§49)	exact symmetry; scaling	queued
Conserved near-miss rule is ultraconservative: forgetting bounded by the new task's mistakes, vs $O(\log T)$ for softmax SGD (§50)	bound prior art; consequence new	E23 running
Honest summary: the mathematics used is borrowed (Noether and Ward identities, Perron-Frobenius and topical maps, Birkhoff contraction, Gibbs/Landauer identities, two-timescale stochastic approximation, mean-field propagation). The applications to race networks were not found in prior work in a few targeted searches (THEORY §32), which is not a claim of priority.		

Theory: collapsing futures, trees of histories, repair

At any moment the network holds a pool of pending futures: each unfired node's projected firing time. A firing collapses the pool: one future becomes history and inhibition cancels its competitors, each leaving a residue (how close it came). Because the order of events depends on the weights, the network's computation is a **tree of possible histories**, and a small weight change can move it to another branch. In a dense network the tree has one branch, so the derivative along it is the whole story. Here it is not.

collapses along the realised history (solid); close calls fork (dashed)

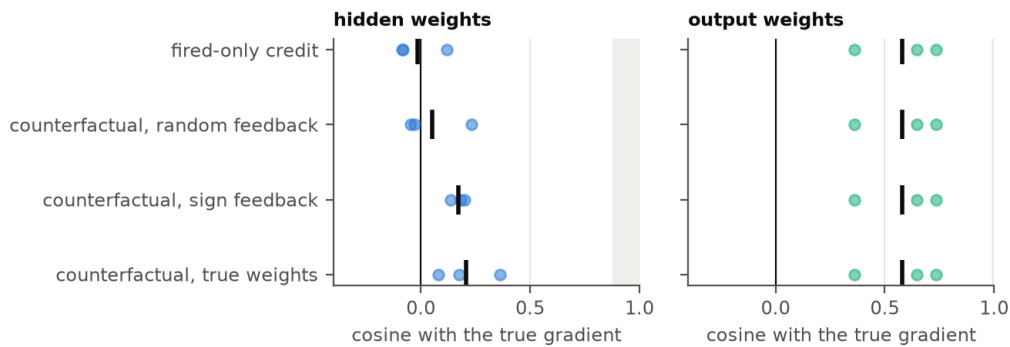


greedy backprop: slope at the solid path only · holistic: sum over forks weighted by p · repair: cheapest fork that is right

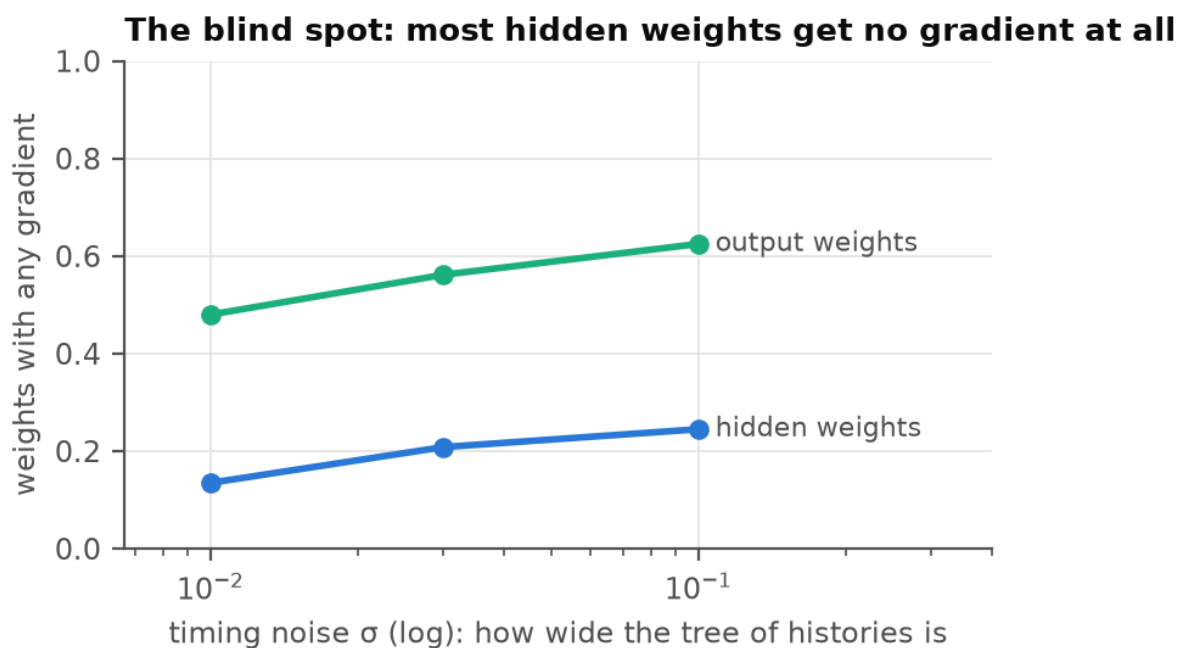
- **Greedy backprop** (EventProp in our setting) follows the realised branch only, and misses how the weights move the forks.
- **Holistic backprop** sums over the close calls, weighted by their probability. Branches are cheap here: a fork changes only events downstream of it, and cancellation ends it early. They can run asynchronously as *shadow events* in the same event queue.
- **History repair** takes the cheapest branch that gives the right answer and changes just that tie. Credit becomes a causal question (would the outcome differ but for this event?) rather than a sensitivity.
- Known ingredients (EventProp, perturbed argmax, surrogate gradients, race logic, Madaline Rule II, spike discontinuity estimation) are credited in experiments/RELATED_WORK.md; the tree-of-histories reading and cancellation residues as its sufficient statistic were not found in prior work.

Checking the theory

M3 · does the update point downhill? (3 seeds; bar = mean; shaded = beyond the estimate's reliability)



M3 measures, on a small network, whether each rule's update points along the true gradient of expected error (estimated by finite differences under timing noise). The output rule does; hidden credit barely does, for every feedback type.

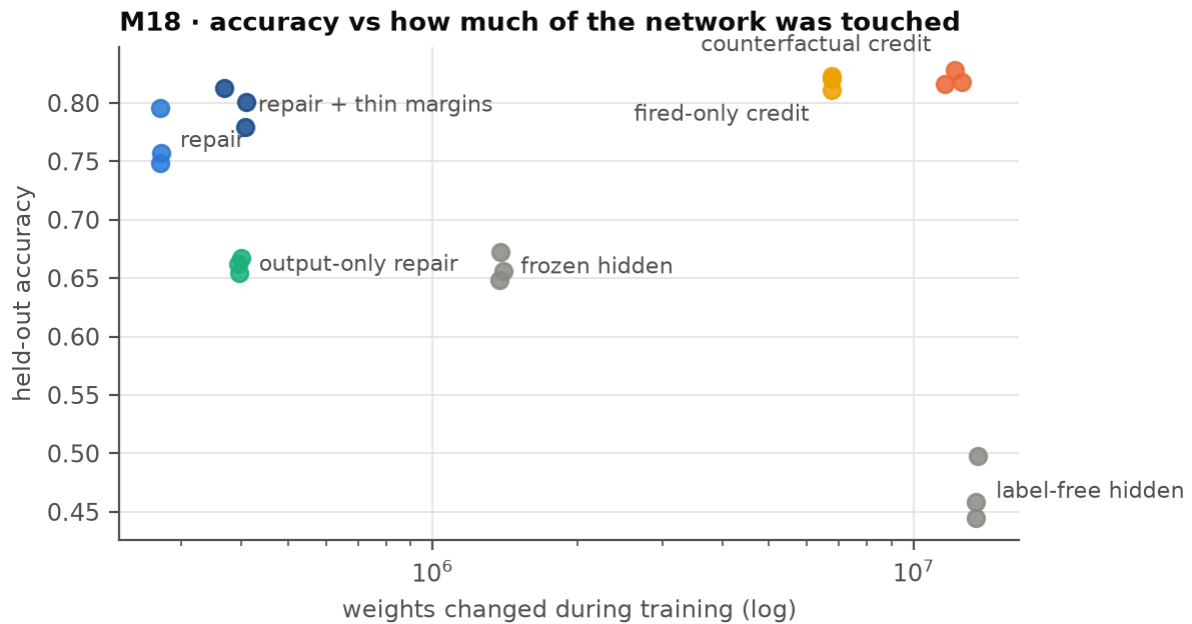


Most hidden weights have no gradient at all along the realised history. Widening the tree (more timing noise) enlarges the set that gets any signal: the blind spot that greedy backprop cannot see.

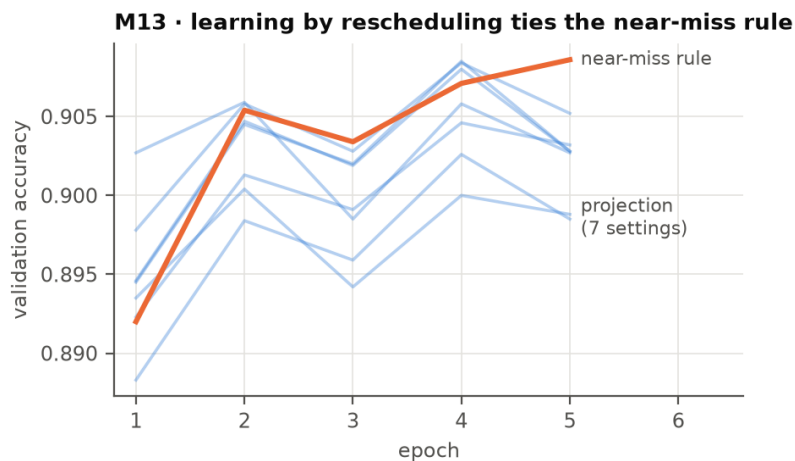
A correction

From M3 we first concluded that the hidden layer learns mainly by its own competition, not from labels. M21 contradicted this: a label-free competitive hidden layer ends far below a frozen random one (0.47 vs 0.66), while label-driven hidden credit adds 16 points. Labels matter to the hidden layer even though their alignment with the gradient of expected error is weak, an open puzzle we are now testing.

New learning rules



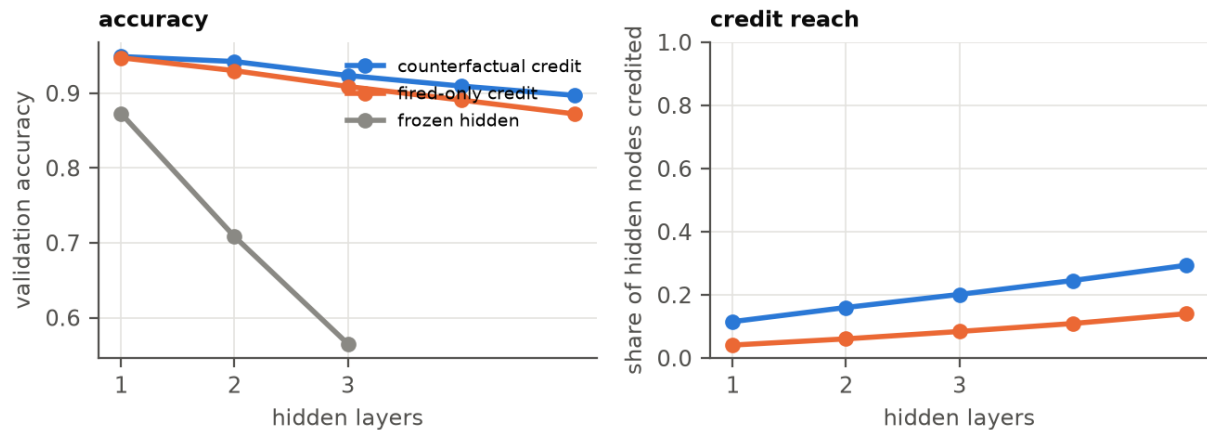
History repair on a small network (14×14 MNIST, 60 hidden nodes, 3 seeds): each mistake is fixed by the cheapest verified single-event change, either rescheduling the output or one hidden node. With thin-margin repairs it reaches 0.80 against 0.82 for the gradient-like rules, while changing 31× fewer weights. Label-free hidden learning (grey, bottom right) is the worst of all.



Learning by rescheduling: “fires by time c ” is a half-space in the weights, so a mistake can be fixed by an exact projection with no learning rate. It ties the near-miss rule: a valid reformulation, not an improvement, and it is now the geometric core of history repair.

Depth: where counterfactual credit pays (E14)

E14 · the advantage of counterfactual credit grows with depth (2 seeds)



Hidden credit per layer through fixed random feedback. With one hidden layer, counterfactual and fired-only credit tie. Deeper, a near-miss node matters only through nodes its spike would have tipped over threshold: a path made of events that did not happen, which only counterfactual credit reaches. At full training the gap is modest (about 2 points at depth 3) but grows steadily with depth, while a frozen hidden stack collapses.

Reinforcement learning: reward only (E13a)

In a first small test, crediting the near misses when wrong reached 0.30, against 0.19 for reward-modulated (winner only) learning and 0.59 with labels. Full runs are queued.

E7 · a learner that lives in a causal stream

Everything above trained the way clocked hardware likes: minibatches, IID epochs, a global learning-rate schedule, a label for every sample, a separate test phase. None of that is needed by an event-driven learner; it is inherited. E7 drops it. The guiding rule is not biological fidelity but this: **where we deviate from biology only because of synchronous-hardware habits, we do not carry the deviation over.**

frames (views of one image) → dark = label arrives orange bar = episode reset



slow state carried across frames: prior · tags · carried label · continuity · homeostasis · consolidation

fast state (one race per frame) restarts at every frame, so the exact solver still applies

each frame is predicted before it may be learned from (prequential); no epochs, no batches

habit	origin	in E7
minibatch updates	GPU throughput	one update per frame (Stage 0 checks this changes nothing)
epochs, reshuffling	batched SGD	one pass over a stream
global learning-rate decay	SGD convergence	per-synapse consolidation
homeostasis from batch means	vectorised statistic	per node, every frame, label or not
dense feedback matrix	cheap matmul	counted as events to eligible nodes only
train phase, then test	benchmark protocol	predict every frame before learning from it
a label for every sample	benchmark convention	10% labels, late labels, or labels the learner asks for
frames with a global $t = 0$	datasets are frames	kept as episode resets; slow state crosses them
latency-coded dense images	MNIST is a frame	kept as a benchmark, not a sensor model

Hypotheses (preregistered)

- **H0** one frame per update learns as well as minibatches of 32.
- **H1** a carried prior cuts the input events needed to decide on later views, without hurting first views.
- **H2** at 10% labels, carried labels and continuity help on real episodes and not on the **shuffled-time control** (same frames, order destroyed).
- **H3** labels the learner asks for (when its race is uncertain) beat random labels on the same budget.
- **H4** synaptic tags make labels that arrive 2 frames late usable.
- **H5** on class-blocked streams the race forgets less than an MLP trained by SGD; replay is reported alongside and may beat both.

Status. 13 E7 result files so far. Pilots use the validation split to set thresholds, which are written into the preregistration before 5-seed confirmatory runs on the test set.

E17 · a continually learning race on a live market stream

A test of the model class on real, non-stationary, asynchronous event data: Binance BTCUSDT trades (millisecond timestamps), 7 pilot days and 21 confirmatory days, one stream. This is a crypto market, not a stock market: freely available stock tick data with raw timestamps was not found. Every 10 s the race restarts, trades stream in as spikes (side × size × tick direction), and the network commits at its first output crossing, or abstains. The label is the price move over the next 10 s *from the moment it decided*. Look-ahead is impossible by construction: a decision uses only trades that already arrived, and weights are taught only with labels revealed before the prediction. **Not a trading system; no live trading.**

- **Learners:** continual race; the same race frozen after the pilot days; a three-output race (up / down / hold) that learns *whether* a move will pay the 2 bp cost, i.e. when to trade; online logistic regression deciding at the end of each window; momentum.
- **Decision rules (preregistered):** competitive if within 1 point of logistic regression while deciding at least 20% earlier; continual learning helps if it beats frozen by 1 point (day-block interval excludes 0); trade selection helps if the hold race's profit proxy beats the others at the same trade fraction. Near 50% for everyone is the expected null for 10 s direction.

learner	accuracy	coverage	profit bp/episode (2 bp)	(10 bp)
b0_d1_w200_s0	0.5913	0.792	-1.442	-7.778
b1_d1_w200_s0	0.5829	1.000	-1.851	-9.851
race_d1_w200_s0	0.5932	0.882	-1.611	-8.670
race_d1_w200_s0_eta0.001	0.5835	0.684	-1.250	-6.725
race_d1_w200_s0_eta0.003	0.5930	0.788	-1.430	-7.733
race_d1_w200_s0_eta0.03	0.5779	0.946	-1.751	-9.316
race_d1_w200_s0_gate	0.5782	0.703	-1.291	-6.914
race_d1_w200_s0_gate_post0.01	0.5888	0.803	-1.466	-7.891
race_d1_w200_s0_post0.001	0.5922	0.800	-1.453	-7.853
race_d1_w200_s0_post0.003	0.5965	0.830	-1.507	-8.148
race_frozen_d1_w200_s0	0.5952	0.782	-1.418	-7.674
race_hold_d1_w200_s0	0.5108	0.014	-0.027	-0.138

Confirmatory days only, prequential. Profit is a diagnostic proxy, not tradable P&L.

- **Preregistered verdicts:** competitive (met: 33% earlier decisions); nominally better than logistic regression (+1.0 points [+0.6, +1.5]).
- **A fairness check made after seeing the results overturns “better”:** at the race's coverage, logistic regression is as accurate (+0.1 points [-0.3, +0.5]); the race ties momentum (+0.2 points [-0.3, +0.6]). The edge came from abstaining on hard windows.
- **Continual learning did not help:** continual – frozen -0.2 points [-0.6, +0.1].
- **Learning when to trade:** the hold race traded 1.4% of episodes, at chance accuracy; its profit ties logistic regression's most confident trades at the same rate. No trade-selection skill.
- **Every learner loses money after costs.** Direction on moves of at least 1 bp is predictable at about 59% (above the ~50% null the preregistration expected; checked for look-ahead), not enough to pay even a 2 bp cost.

Computing with time (E24-E30, 26-27 September)

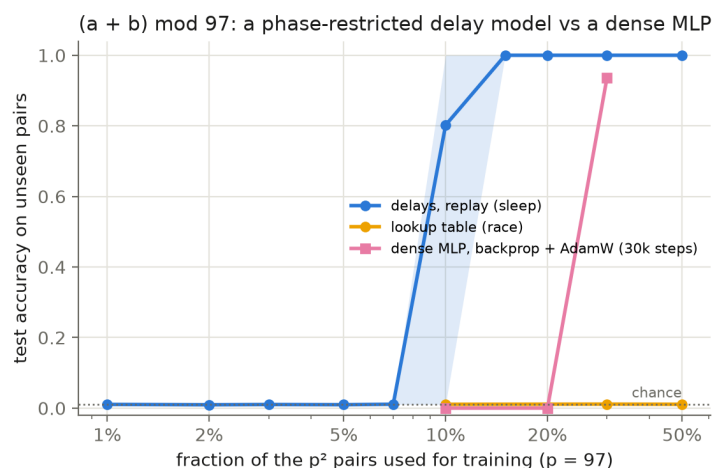
Until E24 the learned part of every race network was synaptic weights; time was a code, never a resource. These experiments ask what time itself computes and how a system can learn it natively: every mechanism an event handler with local state, cost proportional to events, no batches, epochs, replay or global sums. Dense rescues are kept only as diagnostics.

E24: grokking (complete)

$(a + b) \bmod 31$, half the pairs. Every local race variant, with or without sleep downscaling, memorizes: train ≈ 1.0 , test ≤ 0.008 (chance 0.032, 17 runs). Dense MLPs: backprop 0.87, Kolen-Pollack 0.956, feedback alignment 0.000. At $p = 97$ backprop fails at 10% and 20% of pairs and groks at 30% (0.937).

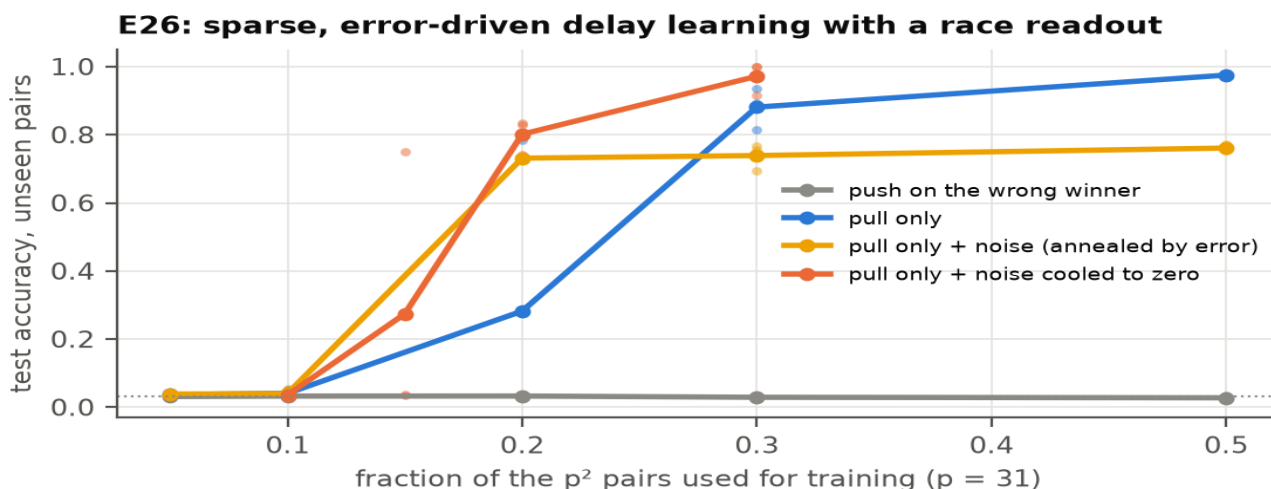
E25: delays instead of lookup

A ring of relays computes $(a + b) \bmod p$ for all pairs with no learning: $4p$ synapses, about $2p$ events per query. Delays learned as phases generalize to 1.000 on unseen pairs from 15–20% of pairs (5 seeds, $p = 31$ –97), including $a \cdot b \bmod p$ (the delays find the discrete logarithm); $a^2 + ab + b^2$ and random tables are out of reach. **This is restriction, not grokking** (§58): the single-phase readout can only express one-character relations, and the learner (replay) is dense.



E26: in a race, winning is positional

The same ring, learned natively: updates only on errors, only on the delays involved, a race readout. Pushing the wrong winner later collapses every detector onto one phase (chance in all 15 runs, $\sim 386k$ updates each). Pulling only the teacher learns the relation ($p = 31$: 0.97–0.98 at 50% of pairs; $p = 59$: 0.95) with $\sim 40k$ updates. Timing noise makes generalization at 20% reliable; cooled to zero it also lifts the accuracy cap.



E27: false positives are specialized, never displaced

false positives handled by	test (5 seeds)	updates
veto (inhibition)	0.914 (0.887–0.934)	15–20k
none (no veto synapses)	0.896 (0.881–0.918)	18–23k
push (delays lengthened)	0.195 (0.18–0.20)	~155k

Patterns “B within Δ after A unless C”. Displacement is destructive; veto helps only a little, because aligning by delay destroys the interval the veto must see (§61: tolerance by holding a long PSP keeps it).

E28: depth needs routing credit, and does not pay yet

arm (hierarchical motifs)	6 classes, 3 seeds	15 classes from 6 motifs, 5 seeds, 60k
depth 1	0.68	0.51
depth 2, counterfactual routing credit	0.72–0.74	0.44
depth 2, critical-path credit only	0.17 (chance 0.14)	–

A race is a routing network: without counterfactual credit (runners-up or cancelled near-misses) depth stays at chance. With it, depth is learnable but not better. Receptive fields show the hidden nodes do learn parts (86 of 124 draw both strongest inputs from one motif), but the readout fails at conjunction: 56% of depth-2 answers are a class sharing one motif with the true one, and none are order errors; synapse caps, windowed readouts and global competition did not fix it (§62). Getting here required §60: pull-only on weights needs a conserved per-node budget in fractional steps, conserving weakening, per-node prices, and enough hidden nodes for Cover's capacity bound (depth 1 rose from 0.29 to 0.68).

E34: depth by composition in time

hierarchical motifs, 15 classes from 6 motifs, 5 seeds	test
depth 2, hold/trigger chains, part window [0, 1.5]	0.97 (0.92–0.99)
depth 2, generic window bank {1, 2, 4}	0.86 (0.79–0.96)
depth 2, single wide window 4	0.79 (0.68–0.93)
depth 1, same nodes on channels	0.39 (0.32–0.48)
E28 (accumulating readout), depth 1 / 2	0.51 / 0.44

Parts are directional hold nodes on channel pairs; a class node holds one part's spike and is triggered by another's, so a single part cannot satisfy it: conjunction and order by construction (§64). Routing is learned by counterfactual pulls under conserved budgets. Learning the part windows from class routes failed (0.08–0.13): a shared part must not be shaped by one class.

E35: nothing given

E27's detectors were told their two channels. With learned hold, trigger and veto weights over all channels and learned hold durations: 1.000 on four seeds and 0.9995 on the fifth, 7.5 synaptic events per episode, 443–1,530 updates in 200k episodes. This is the headline result.

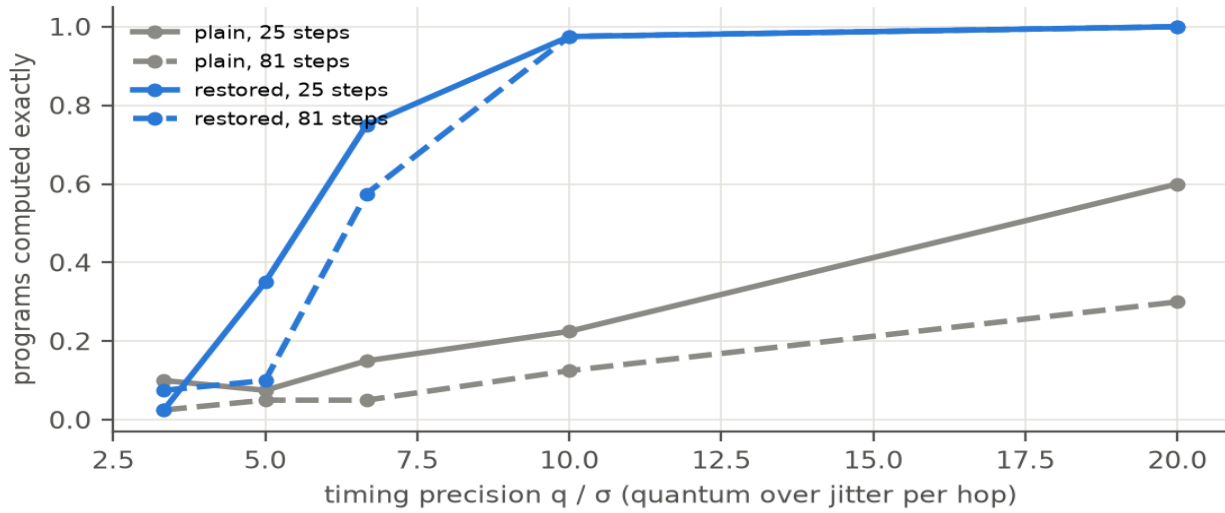
E29: true grokking test, not passed

A general race network that can memorize ($p = n/\text{params} \approx 0.006$), given recurrent delay loops, on E24's encoding. Frozen random loops only memorize (test 0.015); with hidden learning, training collapses whether loop periods are learned or fixed. No grokking yet.

E30: the operator basis is Turing-complete, and restoration makes it scale

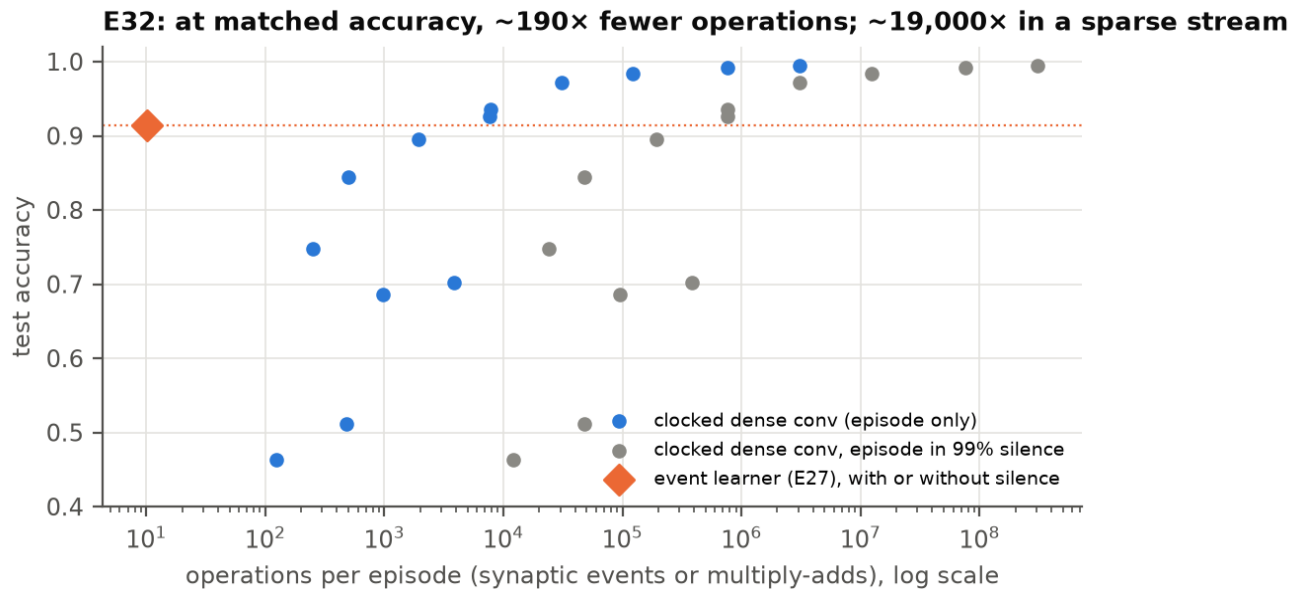
A two-counter Minsky machine as a netlist of Delay, Or, And (a PSP window per input) and Veto nodes plus one reference oscillator; counters are phases of spikes in hold loops. Exact on every test program. Under timing jitter, one comb coincidence per counter per cycle makes reliability independent of program length: precision is the tape, and restoration is its price.

E30: a Minsky machine wired from delay, or, and, veto + one oscillator



E32: a measured frontier against a clocked dense model

E27's task, against a 1-D temporal conv net on binned spikes trained by backprop on the same 200k episodes. The cheapest dense model at the event learner's accuracy (≈ 0.91) needs 1.9k multiply-adds per episode; the event learner 10.2 synaptic events ($\approx 190\times$ fewer operations, $\approx 80\times$ less energy at published per-operation costs). In 99% silence the clocked cost grows $100\times$ and the event cost does not ($\approx 19,000\times$). Training: $\approx 6,000\times$ fewer operations. **Caveats:** the dense model reaches 0.995 at 3M multiply-adds, which the event learner cannot yet buy; and most of the gap is the clock: a sparse, event-driven implementation of the same conv would cost ≈ 106 multiply-adds, only $\approx 10\times$ more (§63).



Theory added (THEORY §53–§63)

- **Clockless means shift-equivariant (§56).** A network of delays, first-ofs, coincidences and vetoes commutes with time shifts, so it cannot add two times, only compare them. One oscillator reference breaks the symmetry to shifts by a period and makes time a cyclic group: exactly one character, which is E25's learnable class. The space-time algebra itself is prior art (Smith 2018; race logic).
- **Credit follows one causal chain (§56.4).** A spike's time depends on one critical path, with derivative 1 along it: exact credit is sparse by construction and does not contract with depth. Routing choices need counterfactuals on top (§57); cancelled near-misses supply them without extra events.
- **Teach the event that should have won (§54, §56.5, §60).** Never displace a loser in time; remove false positives by veto where codes are specific. For timing parameters pull-only suffices; for weights, pull under a conserved budget with prices.
- **Aligning destroys the interval, holding keeps it (§61).** Delay says where in time an event acts, PSP duration how long a node remembers it; veto and ordering need duration.
- **What counts as generalization (§58).** Restriction by the model class, forced generalization above capacity, and grokking (the relation reached while memorizers exist) are different claims; report $p = n / \text{params}$ with every result.
- **Where supremacy can live (§55).** Not in operation counts for static functions (an encoding effect). On the input side the factor is $1 / (\text{spikes per channel per precision bin})$: SHD gives only $\sim 6\times$ at 10 ms bins, so the benchmark must be far sparser in time.
- **Completeness and restoration (§59).** {delay, first-of, coincidence, veto, hold} + one reference is Turing-complete (Minsky; prior art for spiking nets: Maass 1996); a random-walk model with ~ 6 jittered hops per cycle fits the unrestored failures.

Lessons learned along the way

- **Compute discipline.** Running two heavy jobs at once hung the host three times (no swap, and no memory limit on the container; two hard reboots corrupted the filesystem). The third time, the queue ran one job while an ad-hoc debug run ran beside it. Now every computation, debug snippets included, goes through the one-job queue (watchdog at 6 GB free, checked every second), and the container gets hard memory and CPU caps.
- **The queue is not optional (26 September).** A fourth hang: ten ad-hoc jobs beside two queue runners. All runs now go through `queue/run_safe.sh`: one job under a global lock, one thread, a memory watchdog.
- **Dense machinery creeps in.** Replay, phasor sums, offline restart selection and dense updates each rescued a result by leaving the event-driven world; such results are diagnostics, not the direction.
- **Check what a structure gives away.** E25 generalized because its readout could express nothing else.
- **Debug leads reverse.** The shadow neuron led by 5 points in 1-epoch runs and trailed by 0.8 at full length; credit conservation's +8-10 became +1-1.7. Short runs are reported as leads only.
- **Theory can be wrong in informative ways.** A predicted input-redundancy pyramid was refuted by measuring the data (entropy exponent 0.95, not 0.5-0.8); an early smoke test contradicts the predicted size of the activity mode. Both are recorded next to the claims they test.
- **Hidden batch assumptions.** E6 ran homeostasis inside the teaching step, which silently assumed every sample is taught. With scarce labels that would have switched homeostasis off. Batch-derived constants (homeostasis rate) also have to be rescaled when moving to one frame per update.
- **A case that batching hides.** A frame where no hidden node fires crashed the per-frame learner; in batches, some other sample always fired.
- **Undercounted training work.** The feedback fan-out and homeostasis updates were not counted. They are now, as events to eligible nodes only.
- **Controls before claims.** Round 3 looked like a win for counterfactual hidden credit until the fired-only ablation matched it. The single-layer and shuffled-time controls exist for the same reason.

Where this could go

- **Parts, not wholes (§62):** find the pressure that makes hidden nodes detect reusable parts, so that depth pays under native credit; then the true grokking test (E29) with that hidden layer.
- **The complete §60 readout in the main race** (conserved budgets, prices, non-leaky outputs) on SHD.
- **The supremacy benchmark (§55):** a stream with spikes per channel per precision bin $\ll 0.01$, measured in events, latency and updates against dense models given the same priors.
- **Make depth pay:** the theory's three remedies, each with a queued test: centre credit in time coordinates (Ward identity), run the prices on the faster timescale, and use sparse fan-in with enough winners ($k \cdot F \geq G$) or topographic codes to damp pattern chaos.
- **Decide better, not only faster:** relative (MSPRT) stopping via a shared free-energy inhibition, and onset-referenced inhibition so the network can use the *absence* of expected spikes.
- **Make the hidden layer cheap:** sparse fan-in cut synaptic events 14-22× in pilots; if accuracy holds, the inference-energy verdict flips.
- **Live in time:** E7 (a causal stream with scarce, late labels) and E17 (a real market stream, with the network learning when to act) test the asynchronous, continual side of the proposal.
- **Seeds:** 3-5 seeds for every headline number before any claim.

Reproduce: `python report/make_pdf.py` rebuilds this document from `experiments/results` and `report/data.json`.
Preregistrations: `experiments/E*_PREREGISTRATION.md`.